Criminal Peace: The Impact of Organized Crime Hegemony on Homicide Reduction in Brazil

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Abstract

This paper examines the causal impact of criminal governance on homicide rates by exploiting the consolidation of the Primeiro Comando da Capital (PCC) in Ribeirão Preto, Brazil, following a coordinated prison riot in March 2001. Using a sharp regression discontinuity design and a novel data set based on homicides reported by newspaper classified by motive, we find that PCC's rise to hegemony led to a sharp and statistically significant 45% decline in the homicide rate. The reduction is concentrated in killings classified as disputes, with no significant change in interpersonal or robbery and police-related homicides. Placebo tests using other municipalities and non-crime-related deaths confirm that the captured effect is driven by PCC consolidation. Our findings provide causal evidence that hegemonic criminal organizations can reduce certain forms of violence by imposing internal order.

Keywords: Organized crime, Homicides, Violence, Regression Discontinuity Design, Latin America, PCC

JEL Codes: K42, O17, D74, H83

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1 Introduction

Organized crime is a widespread phenomenon in Latin American countries with negative consequences for their populations. Although there are differences in organization and practices in different countries (Lessing, 2022), the presence of organized crime, mainly linked to drug traffic activities, is significant in countries like Brazil and Mexico (Dell, 2015). Meanwhile, Latin America has the highest homicide rates in the world, with deaths related to organized crime accounting for about 50% of the total homicides in the region in 2021 (UNODC, 2023). The relationship between organized crime activities and homicides has been the focus of some studies (e.g., Sobrino, 2022; Biderman et al., 2019; Justus et al., 2018), but identifying causal effects remains a challenge.

While criminal groups are typically associated with the escalation of violence, recent scholarship has proposed that hegemonic criminal organizations can also contribute to the reduction of homicides by imposing internal order and rules among illicit actors. In particular, groups such as the *Primeiro Comando da Capital* (PCC) in Brazil have established systems of internal justice, enforced codes of conduct, and regulation of the illegal drug market (Feltran, 2018; Lessing and Willis, 2019). By deterring internal disputes and violent competition over territory, hegemonic criminal governance can suppress certain types of violence, particularly those linked to turf wars between rival factions (Bertolai and Scorzafave, 2021). However, the scarce evidence of empirical evidence on this pacifying effect remains causally ambiguous.

To address this gap, we estimate the causal effect of PCC's consolidation of power on homicide rates in Ribeirão Preto, São Paulo, following a prison mega-rebellion in March 2001, using a regression discontinuity design (RDD). Our analysis is based on a unique dataset constructed from crime reports published by a local newspaper, allowing us to classify homicides by motive. We complement our main analysis with placebo tests using other municipalities and falsification outcomes to assess whether the observed decrease in homicides can be attributed to broader trends or data anomalies.

Our results show a sharp and statistically significant decline in homicides after PCC's consolidation, specifically concentrated in killings classified as disputes (score-settling, revenge, fights, and others). We estimate a reduction of 3.5 homicides per 100,000 inhabitants, equivalent to a 45% drop, after the 3rd bimester of 2001, with no significant change in interpersonal or robbery and police-related homicides. Placebo analyses in other municipalities without PCC hegemony and in non-crime-related mortality outcomes (e.g., cancer deaths) reveal no similar discontinuities.

Our study makes three main contributions to the literature. First, it provides causal evidence that hegemonic criminal governance can reduce violence, using a quasi-experimental design. Second, we study the dynamic transition from decentralized violence to coordinated criminal control, rather than marginal variation in PCC hegemonic power. Third, we clarify the scope of criminal governance effects on different types of homicides, by showing that the decline in violence is concentrated in killings related to organized crime disputes.

The remainder of the paper is organized as follows. Section 2 provides background information on the PCC and its expansion. Section 3 reviews the related literature. Section 4 describes the empirical strategy and data. Section 5 presents the main results and the placebo tests. Section 6 discusses the findings and their broader implications.

2 Organized Crime in Brazil

2.1 PCC and its consolidation

The PCC emerged in the early 1990s in the prison system of São Paulo, Brazil's largest and richest state. Like other prison gangs, it was initially formed to improve inmate conditions and provide protection against rival groups (Biondi, 2018). An increase in incarceration rates facilitated PCC's consolidation: between 1993 and 2019, the prison population of São Paulo more than quadrupled, while prison capacity did not keep pace. Overcrowding expanded the recruitment pool and weakened institutional control. In this environment, controlling prison life increased the value of gang membership, strengthening PCC's dominance (Lessing, 2016).

The PCC was founded in 1993 in Taubaté, a medium-sized city near São Paulo, in one of the most violent prisons of the state. The group gained control after eight inmates killed two rivals and threatened others following a soccer match (Dias, 2011). Its early expansion was marked by violent repression of rivals and strict internal discipline (Stahlberg, 2021).

On February 18, 2001, the PCC coordinated rebellions in 29 prisons spread over São Paulo state, covering roughly 35% of the state's facilities. Although triggered by the transfer of PCC leaders, the underlying objective was to restructure power within the prison system. The scale of the uprising, resulting in approximately 20 inmate deaths with bodies publicly displayed, forced authorities to acknowledge the influence of the group for the first time (Dias and Darke, 2016). The rebellion expanded beyond the central prisons of São Paulo. In Ribeirão Preto (located 200 miles from the capital), an uprising occurred on March 29, 2001, motivated by two primary objectives: first, to retaliate against the failed escape of the local PCC leader two days prior; and second, to distract prison guards and facilitate the targeted execution of rival factions of PCC (Folha de S.Paulo, 2001c).

The 2001 coordinated prison riots consolidated the PCC's power by demonstrating the strength and ability of the organization to coordinate actions in multiple prisons, leading to (a) the recruitment of new members, (b) the establishment of control over additional prisons in the state of São Paulo, and (c) the official recognition of public authorities (Ferreira, 2019) . We argue that the consolidation of the PCC in Ribeirão Preto occurred in the months following the mega-riot of March 2001. In the immediate aftermath of the riot, which demonstrated the organizational capacity and influence of PCC, public events were attributed to the group (Folha de S.Paulo, 2001a), confirming its growing presence and control. By August 2001, reports indicated that the PCC had established a form of "peace" in Ribeirão Preto prison (Folha de S.Paulo, 2001b), further solidifying its dominance in the region.

After 2006, PCC entered a new era of expansion and consolidation. In May 2006, they organized and executed an extraordinary demonstration of power. The PCC launched a synchronized wave of attacks, which would become known as *Crimes de Maio* (Crimes of May), in response to a decision by state authorities to transfer more than 700 inmates, including the group leader, to a maximum security prison far from the capital. In 2006, PCC coordinated simultaneous riots in 74 prisons around the state and launched hundreds of attacks on police, firefighters, civilians, and urban infrastructure. In 2025, PCC operates in almost all 27 Brazilian states and maintains an estimated 100,000 members throughout the country, consolidating its dominance in both prison and street-level criminal activities (UK Home Office, 2025)

2.2 How an organized crime hegemony reduces crime?

Feltran (2018) presents two main explanations for how PCC hegemony leads to a reduction in crime. First, PCC establishes its own system of rules and justice, enforced through its statute. These rules include prohibitions against killing without prior organizational approval, stealing from other members, and engaging in romantic relationships with the partners of members. The PCC maintains a justice committee that debates and resolves disputes.

Second, the PCC regulates the illegal market. For example, the retail sale of drugs in São Paulo is subject to price controls set by the organization, which reduces competition between drug sales points and minimizes conflicts. Similarly, the PCC exerts other criminal activities (such as car thefts), imposing order, and reducing violence.

Importantly, Feltran (2018) argues that the PCC does not have a monopoly on the illegal market in São Paulo but rather maintains hegemony. This hegemony exists because most crime participants are either directly affiliated with the PCC or coerced into compliance by the organization. Although nearly all drug points in São Paulo are controlled by the PCC or by individuals who align with its values (even if not formally a member of the PCC), it is not a monopoly. According to Feltran (2018), small independent drug dealers that are not under PCC control can still sell drugs at different prices. However, these dealers will never grow significantly unless they comply with the PCC rules.

Bertolai and Scorzafave (2021) using a theoretical model, explain the mechanisms through which a hegemonic prison gang can reduce violence. In the absence of such a hegemonic group, drug dealers use violence to compete for territory and consumers, since legal enforcement is not available. When a single prison gang consolidates control over the prison system, it can influence the behavior of dealers outside the prison by conditioning the inmates' welfare on compliance with its rules. Dealers anticipate future incarceration and prefer to follow the gang's directive in exchange for lenient treatment in prison. The gang's ability to promise rewards for compliance and impose harsh punishment for defection allows it to enforce a collusive equilibrium and a reduction in homicides associated with turf wars.

3 Related Literature

One branch of the literature has examined the determinants of homicide dynamics in Brazil, attributing the variation in homicide rates to structural changes such as demographic changes (Cerqueira and Moura, 2016), improvements in labor market conditions (dos Santos and Kassouf, 2013), the enactment of stricter alcohol control policies, such as the 'Lei Seca' (Biderman et al., 2009), and reduction in firearms circulation (Cerqueira, 2014).

Other branch of the literature study especially the role of organized crime in shaping local patterns of violence. Justus et al. (2018) find no evidence that PCC played a significant role: using the intensity of the 2006 prison riots and attacks as a proxy for the territorial influence of the gang, they show that areas more exposed to PCC activity did not experience larger reductions in homicide. There are two conceptual distinctions between their study and ours. First, they examine the magnitude of PCC power across municipalities in 2006, when the organization was already consolidated throughout the state. We focus on the earlier stage of its expansion, beginning in 2001, which captures the process of consolidation. Second, their identification strategy captures marginal variation in PCC power, while our analysis focuses on a dichotomous shift, whether or not the gang consolidated its influence. It is plausible that municipalities with low, moderate or high levels of PCC presence all experience similar reductions in homicides, as they equally benefit from the internal justice system imposed by the gang. This could lead to limited variation in outcomes across different levels of exposure.

In contrast to the previous study, Biderman et al. (2019) provide evidence that PCC entry into *favelas* in the city of São Paulo between 2005 and 2009 led to a reduction in homicides within these territories. To identify the presence of PCC, the authors used data from an anonymous crime hotline, selecting calls that mentioned PCC and were from *favelas*. They find that PCC entry reduced violent crime, with no significant effect on property crime, and that the impact was more pronounced in areas with higher initial levels of violence. Our study extends these findings in two ways: (1) rather than examining the presence in a specific geographical area (*favelas*), we investigate the effects of PCC consolidation in one entire municipality; and (2) we disaggregate homicide data to explore which specific types of killing are most affected by PCC expansion.

Finally, using data from the state of São Paulo, Bertolai and Scorzafave (2021) define PCC domination based on prison rebellion records since 1993, identifying whether the gang controlled at least one prison within a given micro-region. They show that before PCC hegemony, drug trafficking and homicide rates were positively correlated. However, once the PCC established territorial control, this correlation is not significant, suggesting that the gang's regulatory mechanisms reduced violence. They also show that PCC domination remained stable after 2001, covering approximately 40% of municipalities in the state.

4 Methodology & Data

4.1 Empirical Strategy

We use regression discontinuity design (RDD) where: for each event i, there is a dependent variable Y_i and a continuous time variable T_i, where the definition of treated or control events depends on the location of T_i. Here, Y_i refers to the homicide rate per 100,000 inhabitants for each period of two months (bimester), and T_i refers to the bimester periods, ranging from 1998 to 2005. We restrict the analysis to 2005 to capture the effects of the PCC consolidation phase, rather than the post-2006 period when the organization was already consolidated in the state of São Paulo and began expanding to other states and internationally.

We apply a sharp RDD, where participation in the treatment is a deterministic function. The probability of treatment is equal to 1 after the cut-off and 0 before. It is assumed that the homicide rates per 100,000 inhabitants have two potential outcomes, $Y_i(1)$ and $Y_i(0)$, corresponding to the outcomes under treated and control conditions, respectively. Thus, we have:

$$Y_i(0) \text{ if } \quad T_i < c \tag{1}$$

$$Y_i(1) \text{ if } \quad T_i \geq c \tag{2}$$

Our threshold c for our baseline estimations are the 3rd bimester of 2001 (May/June). Although the mega-riot occurred at the end of March, technically within the 2nd bimester, we use the 3rd bimester as the cutoff because we expect any effects on homicide rates to materialize with a short delay. This lag accounts for the time it takes for the PCC to consolidate its control, reorganize local criminal structures, and influence violent dynamics on the ground.

The assumption of this method is that the observed events (i) close to the cutoff point are similar, in unobservable characteristics, to the events just above the cutoff, except for receiving or not receiving the treatment. In this sense, the average treatment effect can be intuitively represented as the distance between:

$$\tau_{SRD} = E[Y_i(1) - Y_i(0) \mid X_i = c] \tag{3}$$

However, this distance cannot be directly observed because we never see both curves for the same value. Thus, the treatment effect is estimated using a sharp RDD regression model defined as follows:

$$Y_i = \tau_{SRD} T_i + \beta_1 (X_i - c) + \beta_2 T_i (X_i - c) + \varepsilon_i \tag{4}$$

where Y_i is the homicide rate per 100,000 inhabitants, X_i is the bimester of the observed rate, c is the cut-off point marking PCC's entry, T_i is a dummy variable equal to 1 if the event occurs after the cutoff point and 0 otherwise, and ε_i is an error term.

We use local linear regressions around the discontinuity to estimate nonparametrically the coefficient of interest, τ_{SRD}[1].

4.2 Data

In this study, we employ a dataset constructed by the Laboratory for Studies and Research in Social Economics (LEPES/USP), based on crime reports published in the newspaper /textitJornal Verdade. This local newspaper from the municipality of Ribeirão Preto covered detailed accounts of homicides in the city and surrounding region from 1991 to 2006.

To compile this information, we developed a digital form that allowed us to convert qualitative narratives from news reports into a structured set of indicators. This transformation enabled us to characterize the dynamics of crime in Ribeirão Preto and construct a detailed dataset.

The dataset includes information on homicides, such as the date, time and location of each event; the motive behind the homicide (e.g., drug trafficking, revenge, gang conflict, police intervention, domestic disputes, robbery, accidents, among others); and victims identification details (name, sex, race, age, occupation, and place of residence and criminal background).

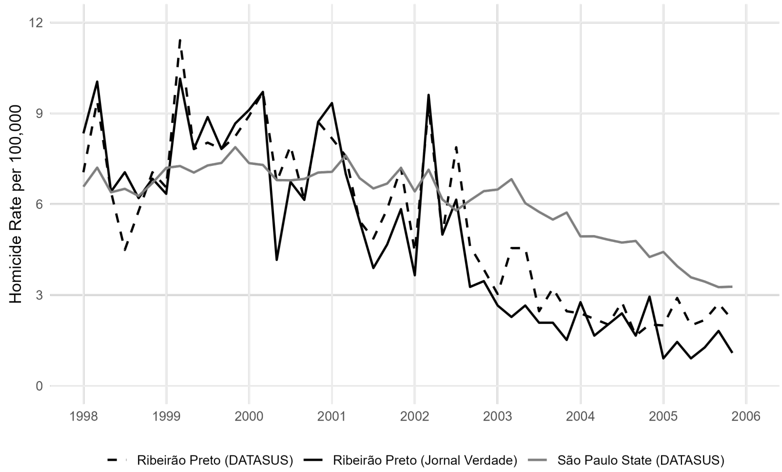
To construct the dataset, we applied a four-stage validation process comprising initial tabulation, second tabulation, review, and final verification. Each stage was conducted by a different individual to minimize subjective bias associated with the interpretation of qualitative sources. Although the data source is not official, we demonstrate in the descriptive section that the number of homicides recorded in our dataset aligns with the official source.

4.3 Descriptives

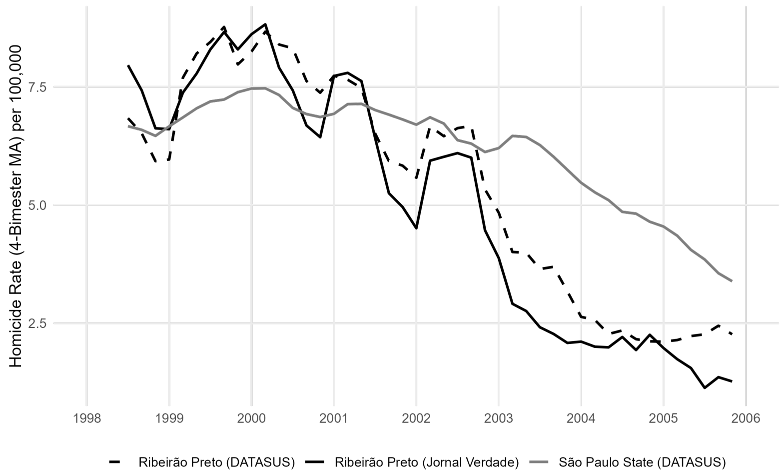
Figure 1a shows the homicide rates in Ribeirão Preto over time, as reported by *Jornal Verdade* and DATASUS. The trajectories of both series are similar. We performed a non-parametric Wilcoxon signed-rank test on the yearly paired data, to assess whether their central tendencies differ. The test yields a statistic of V = 8 and a p-value of 0.1953. Therefore, we fail to reject the null hypothesis, suggesting no statistically significant difference in central tendency.

Figure 1a shows a downward trend in homicide rates after 2001 in both São Paulo State and Ribeirão Preto. This pattern becomes clearer in Figure 1b, which presents the four-brimester moving average and highlights the sustained decline after 2001.

[1]Before proceeding to analysis, we checked the presence of autocorrelation before the cutoff, since it could bias the RDD results. Table 7 shows that the pre-intervention residuals have no significant autocorrelation at any lag length.



(a) Per-Bimester Homicide Rates

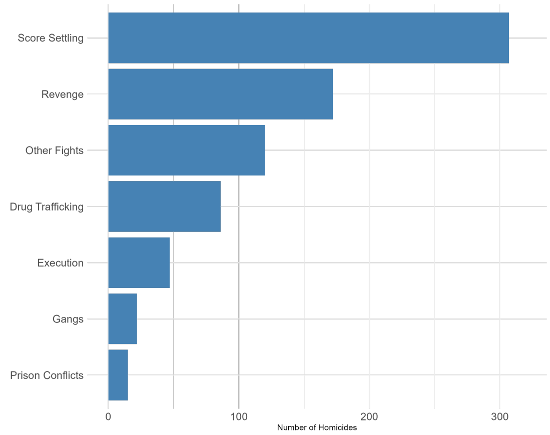


(b) Four-Bimester Moving Average

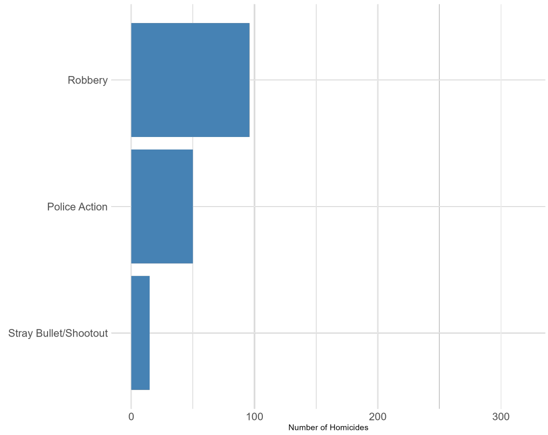
Figure 1: Homicide Rates in Ribeirão Preto and São Paulo

Notes: Panel (a) shows raw homicide rates per 100,000 inhabitants calculated for each bimester. Panel (b) presents the four-bimester moving average. Data sources: DATASUS and *Jornal Verdade*.

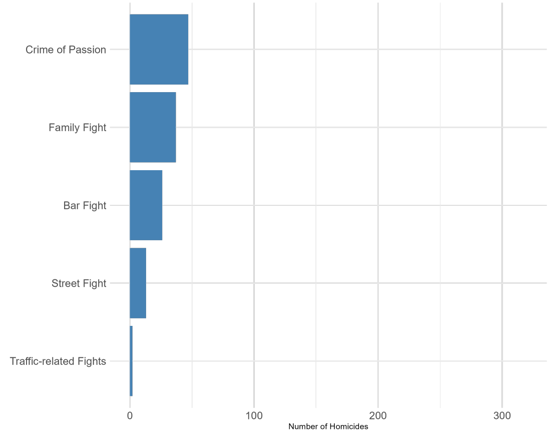
We categorize the homicides into four distinct groups. The first and largest group is Disputes, which includes score settling, revenge, and organized crime-related fights (excluding interpersonal incidents such as bar and family fights). This group also includes drug trafficking, as these cases likely reflect internal conflicts within drug networks. Deaths resulting from police operations related to drug activity are categorized under Police Action, not Disputes. The second group includes robbery, police action, and stray bullet/shootout incidents. Although heterogeneous, these causes reflect both the economic activities typically associated with organized crime (e.g., robbery) and the reactive violence from law enforcement. If the influence of PCC expands, increases in both robbery-related homicides and confrontations with police were expected. The third group, interpersonal violence, encompasses bar fights, family fights, crimes of passion, and similar individual conflicts. The fourth and final group, labeled other, includes accidents, self-defense, and cases with unknown motives.



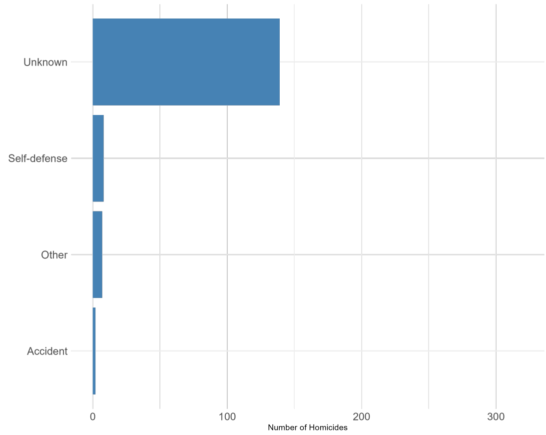
(a) Disputes



(b) Robbery & Police Action



(c) Interpersonal



(d) Others

Figure 2: Occurrence of Crime by Category

Table 1: Mean Homicide Rates per 100,000 Inhabitants, Before and After the 3rd Bimester of 2001

Period	All Homicides	Disputes	Interpersonal	Robbery/Police Action
Before 3rd Bimester 2001	7.77	5.04	0.74	0.96
After 3rd Bimester 2001	3.04	1.86	0.34	0.46

In table 1 we show the average homicide rate before and after the 3rd bimester of 2001. The overall homicide rate fell from 7.77 to 3.04 per 100,000 inhabitants. This decline was driven primarily by a reduction in Disputes-related homicides, which dropped from 5.04 to 1.86 — indicating a significant contraction in violence associated with organized crime conflicts. Interpersonal homicides, Robbery and police-action-related deaths also decreased, though to a lesser extent.

5 Results

5.1 All homicides

Figure 3 shows a sharp decline in the homicide rate around the 3rd bimester cutoff. This pattern is confirmed by the RDD estimates presented in Table 2, where we report results for three potential cutoff points in 2001: the 2nd, 3rd, and 4th bimester. The only statistically significant effect is found in the 3rd bimester (May–June), where we estimate a reduction of 3.5 homicides per 100,000 inhabitants. Given that the average homicide rate before the cut-off was 7.77 (Table 1), this represents a 45% decrease in the rate of homicides.

In the Appendix, we replicate the analysis using official SIM/Datasus mortality data and find comparable results (Table 8). We also estimate the model using different bandwidths and allow for both linear and quadratic polynomial specifications (Table 9). The results are robust across a range of bandwidths, particularly for the linear specification with 6 to 15 bimesters. As the bandwidth increases and the model includes a quadratic polynomial (p=2), the estimates become less precise. However, the estimated effect is negative and statistically significant in most of these specifications, suggesting a substantial reduction in the homicide rate around the third bimester of 2001.

Table 2: RDD: Effects of PCC Hegemony on homicide rate (per 100,000 inhabitants)

Variable	2nd bim/2001	3rd bim/2001	4th bim/2001
Jornal Verdade	-4.119 (2.829)	-3.473** (1.390)	-1.189 (2.695)
Observations Left Side	4	5	5
Observations Right Side	5	6	6
Bandwidth (Bimester)	5	6	6

Note: The table reports local linear RDD estimates. The dependent variable is the homicide rate per 100,000 inhabitants, which we construct using records from *Jornal Verdade*. The running variable is time (bimesters). The cutoff equals the 2nd, 3rd, and 4th bimesters of 2001. We report bias-corrected estimates with robust standard errors in parentheses. The procedure selects MSE-optimal bandwidths, expressed in bimesters. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

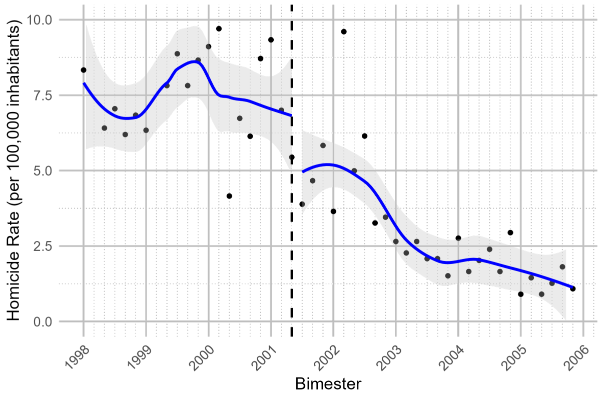


Figure 3: Homicides Rate

Notes: The dashed vertical line represents the 3rd bimester cut-off. The blue line is non-parametric locally weighted regression curves at each point. The shaded gray area represents a 90% confidence interval.

5.2 Type of Homicides

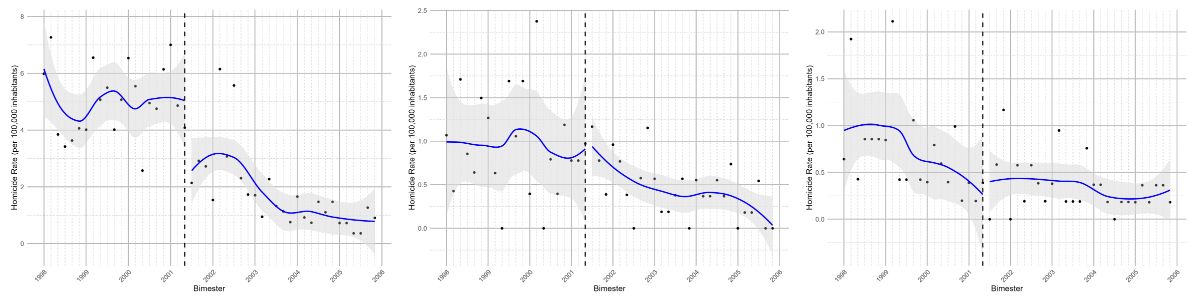
Table 3 shows the impact of PCC by type of crime, using the 3rd bimester cutoff. The largest and most significant effect is observed in crimes classified as Disputes, which include incidents such as score-settling, revenge, execution, gang activity, prison conflicts, drug trafficking, and other fights (including those unrelated to bars, family, and car traffic). This finding indicates that PCC's consolidation of power significantly reduced violent conflicts within criminal organizations. Given that the average rate of homicides by disputes before the cut-off was 5.04 (Table 1), this represents a 65% decrease in the rate of homicides by disputes.

Table 3: RDD: Effects of PCC Hegemony on Homicide Rates

Variable	All	Disputes	Interpersonal	Robbery & Police Action
Coefficient (Std. Error)	-3.473** (1.390)	-3.304** (1.317)	0.217 (0.386)	0.179 (0.205)
Observations Left Side	5	7	6	5
Observations Right Side	6	8	7	6
Bandwidth (Bimester)	6	8	6	6

Notes: The table reports local linear RDD estimates. The running variable is time (bimesters), and the cutoff equals the 3rd bimester of 2001. The dependent variables are homicide rates per 100,000 inhabitants by cause group: all causes combined; disputes (organized disputes, drug trafficking, and other fights); interpersonal; and robbery and police action (robbery, police action, and stray bullet shootout). The dependent variables were constructed using records from *Jornal Verdade*. We report robust standard errors in parentheses. The procedure selects MSE-optimal bandwidths, expressed in bimesters. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

We do not find a statistically significant impact on homicides arising from interpersonal conflicts (e.g., bar fights, family fights, street fights, car-traffic-related fights and crimes of passion). Additionally, the effect is negligible for homicides related to the business of organized crime (e.g., robbery) and its consequences, such as police action and stray bullet/shootout incidents. These non-significant results reinforce the hypothesis that PCC's hegemony primarily influenced internal criminal disputes rather than general violence or violence resulting from criminal enterprises. Figure 4 illustrates the results found in the RDD estimation.



(a) Dispute (b) Robbery & Police Action (c) Interpersonal

Figure 4: RDD Plots by Crime Category: All Causes, Dispute, and Interpersonal.

Notes: The dashed vertical line rpresents the 3rd Bimester Cutoff. The blue line is non-parametric locally weighted regression curves at each point. The shaded gray area represents a 90% confidence interval.

5.3 Placebo

We estimate the same RDD model using data from São José do Rio Preto to assess whether the observed discontinuity in homicide rates is specific to Ribeirão Preto or reflects broader trends (e.g. the implementation of a statewide public security policy[2]). São José do Rio Preto is in the state of São Paulo and had approximately 408,000 inhabitants in 2010 (compared to 605,000 in Ribeirão Preto). São José do Rio Preto did not have a penitentiary facility at the time, suggesting that PCC hegemony was not consolidated there in 2001.

If the drop in homicide rates around the 3rd bimester of 2001 in Ribeirão Preto was driven by the early presence of PCC, we should not observe a similar pattern in São José do Rio Preto. Table 4 supports this hypothesis: the estimated effects between different bimesters are statistically insignificant, not indicating a clear discontinuity in homicide rates.

[2] In Brazil, state governments are constitutionally responsible for ensuring public security.

Table 4: RDD Placebo: Effects of PCC hegemony on São José do Rio Preto Homicide rates (per 100,000 inhabitants)

Variable	2nd bim/2001	3rd bim/2001	4th bim/2001
São José do Rio Preto	-0.436 (0.719)	-1.050 (0.898)	0.015 (1.446)
Observations Left Side	6	6	5
Observations Right Side	7	7	6
Bandwidth (Bimester)	7	7	6

Notes: The table reports local linear RDD estimates. The running variable is time (bimesters). The cutoff equals the 2nd, 3rd, and 4th bimesters of 2001. The dependent variable is the homicide rate per 100,000 inhabitants in São José do Rio Preto, which we construct using administrative mortality records from SIM/Datasus. We report robust standard errors in parentheses. The procedure selects MSE-optimal bandwidths, expressed in bimesters. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

We also performed a similar placebo test using data from Uberlândia, a city located in the neighboring state of Minas Gerais. Uberlândia is demographically and economically comparable to Ribeirão Preto: Uberlândia had approximately 604,000 inhabitants and a GDP per capita of R$31,569 in 2010, while Ribeirão Preto had around 605,000 inhabitants and a GDP per capita of R$30,210 in the same year. Although Uberlândia has a penitentiary, there is no record of any prison rebellion in 2001. Although PCC is the only major criminal faction operating in the state of São Paulo, Minas Gerais is home to more than 13 criminal organizations (Biblioteca de Segurança Pública, 2024). This fragmentation reduces the likelihood of coordinated criminal governance in the state. Consistent with this context, Table 5 shows no evidence of a discontinuity in homicide rates around the 2001 cutoff in Uberlândia.

Table 5: RDD Placebo: Effects of PCC hegemony on Uberlândia Homicide rates (per 100,000 inhabitants)

Variable	2nd bim/2001	3rd bim/2001	4th bim/2001
Uberlândia	-0.540 (0.701)	0.222 (0.388)	-0.077 (0.375)
Observations Left Side	6	4	3
Observations Right Side	7	5	4
Bandwidth (Bimester)	7	4	4

Notes: The table reports local linear RDD estimates. The running variable is time (bimesters). The cutoff equals the 2nd, 3rd, and 4th bimesters of 2001. The dependent variable is the homicide rate per 100,000 inhabitants in Uberlândia, which we construct using administrative mortality records from SIM/Datasus. We report robust standard errors in parentheses. The procedure selects MSE-optimal bandwidths, expressed in bimesters. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Finally, we test whether there is any evidence of a discontinuity in cancer-related deaths in Ribeirão Preto. The purpose of this placebo test is to ensure that broad changes in mortality or in data-recording practices around the cutoff do not account for the observed discontinuity in homicide rates. Cancer deaths, which are not related to criminal activity, serve as a falsification outcome. As shown in Table 6, we find no statistically significant change around the cut-off point.

Table 6: RDD Placebo: Effects of PCC hegemony on neoplasia death rates in Ribeirão Preto (per 100,000 inhabitants)

Variable	2nd bim/2001	3rd bim/2001	4th bim/2001
Ribeirão Preto (Neoplasia)	3.602 (4.981)	1.404 (1.820)	0.657 (0.714)
Observations Left Side	4	7	6
Observations Right Side	5	8	7
Bandwidth (Bimester)	5	8	6

Notes: The table reports local linear RDD estimates. The running variable is time (bimesters). The cutoff equals the 2nd, 3rd, and 4th bimesters of 2001. The dependent variable is neoplasia death rate per 100,000 inhabitants in Ribeirão Preto, which we construct using administrative mortality records from SIM/Datasus. We report robust standard errors in parentheses. The procedure selects MSE-optimal bandwidths, expressed in bimesters. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

6 Discussion

This study provides novel empirical evidence on the relationship between organized crime consolidation and homicide dynamics, using the case of PCC in Ribeirão Preto, São Paulo. Our RDD analysis reveals a substantial and statistically significant reduction in homicide rates following the PCC's consolidation of power around the 3rd bimester of 2001. Specifically, the homicide rate decreased by approximately 45%, with the decline concentrated in dispute-related killings, a category closely tied to criminal organization rivalries and internal score-settling. This pattern supports theoretical arguments suggesting that hegemonic criminal organizations can enforce a form of "criminal governance" that suppresses decentralized violence.

Our disaggregated analysis highlights that the decline in violence was not uniform across all types of homicides. While dispute-related killings dropped by roughly 65%, there was no statistically significant change in interpersonal homicides or those related to robbery and police confrontation. PCC appears to impose internal order by reducing conflict among criminal actors while not necessarily curbing broader interpersonal violence.

The results are robust across multiple specifications, bandwidths, and polynomial orders and are not replicated in placebo tests using other municipalities or outcomes unrelated to criminal activity (e.g., cancer mortality). The absence of discontinuities in São José do Rio Preto and Uberlândia, which are two demographically and economically similar cities without documented PCC dominance in 2001, further strengthens the causal interpretation of our findings.

Our findings contribute to a broader literature on the paradoxical role of organized crime in reducing violence. While criminal groups are typically associated with illicit activities and violence, their internal cohesion and enforcement mechanisms can, under certain conditions, reduce lethal competition. In this regard, our results are consistent with the previous work by Bertolai and Scorzafave (2021), who emphasize how hegemonic groups can establish property rights and regulate behavior within illegal markets, and with Biderman et al. (2019), who find that PCC reduces violent crime in Sao Paulo city. However, our work is different in the sense that we focus on the moment of consolidation, capturing the dynamic transition from fragmented violence to centralized control, and we use motive-specific homicide data.

Nonetheless, we must interpret our findings with caution. The reduction in homicides observed in Ribeirão Preto does not necessarily indicate an improvement in the rule of law or a strengthening of democratic institutions. While the PCC's dominance appears to have curtailed lethal disputes, it does so by establishing its own extralegal justice system, where decisions are made by internal committees and enforced without due process of law. A limitation of the paper is that our empirical strategy allows us to estimate only the short-run impact of the consolidation of PCC. Therefore, we cannot infer the effect of PCC on homicide dynamics after some years of PCC consolidation.

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7 Appendix

Table 7: Ljung-Box Test for Residual Autocorrelation at Different Lags

Lags	Statistic	p-value
4	3.759	0.440
6	4.558	0.602
8	8.243	0.410
10	10.010	0.440
12	13.549	0.330

Notes: The table reports Ljung–Box tests for residual autocorrelation from the pre-intervention ARIMA model. We estimate the model using data before the 3rd bimester of 2001. The null hypothesis states that the residuals show no serial correlation up to the specified lag.

Table 8: RDD: Effects of PCC Hegemony on homicide rate (per 100,000 inhabitants)

Variable	2nd bim/2001	3rd bim/2001	4th bim/2001
SIM/Datasus	-2.050 (1.544)	-2.864*** (0.731)	-0.469 (2.310)
Observations Left Side	5	5	6
Observations Right Side	6	6	7
Bandwidth (Bimester)	5	5	6

Note: The table reports local linear RDD estimates. The dependent variable is the homicide rate per 100,000 inhabitants, which we construct using administrative mortality records from SIM/Datasus. The running variable is time (bimesters). The cutoff equals the 2nd, 3rd, and 4th bimesters of 2001. We report bias-corrected estimates with robust standard errors in parentheses. The procedure selects MSE-optimal bandwidths, expressed in bimesters. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 9: RDD: Effects of PCC Hegemony on homicide rate (per 100,000 inhabitants)

Bandwidth (Bimester)	Polynomial Order	Coefficient (Std. Error)
6	p = 1	-3.473** (1.409)
6	p = 2	0.642 (2.351)
9	p = 1	-3.450*** (1.270)
9	p = 2	-3.988** (1.770)
12	p = 1	-2.361** (1.061)
12	p = 2	-4.399*** (1.684)
15	p = 1	-1.806* (1.027)
15	p = 2	-3.623** (1.475)
18	p = 1	-1.790* (1.022)
18	p = 2	-2.585** (1.262)

Notes: The table reports local RDD estimates for the homicide rate per 100,000 inhabitants. The running variable is time (bimesters), and the cutoff equals the 3rd bimester of 2001. We vary the bandwidth and the polynomial order (p = 1 and p = 2) to assess robustness. We report robust standard errors in parentheses. Bandwidths are in bimesters. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.